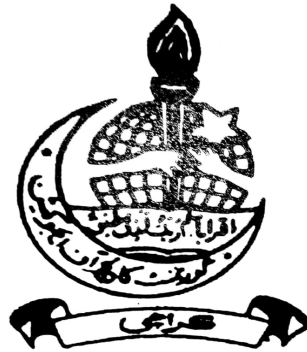


USING DISCREPANT EVENTS IN SCIENCE CLASSROOM TO ENHANCE STUDENTS' ENGAGEMENT AT 6TH GRADE

By

BISMA HAYAT D/O HAYAT MUHAMMAD



SUPERVISED

By

Prof. Tabassum Kausar

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بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

ABSTRACT:

This action research study aimed to explore the effectiveness of utilizing discrepant events in science demonstrations to promote student engagement in within a 6th-grade classroom setting. The study employed action research, to address the challenge of maintaining student interest in science. Embarking on an exciting exploration, this research set out to uncover the magic of science education by introducing science demonstrations through a strategy of discrepant events with technology. Through a hands-on approach and engaging activities, the study aimed to enhance students' curiosity and deepen their understanding of scientific concepts. By observing students' reactions and testing their knowledge before and after the demonstrations, the research sought to measure the impact of these interactive experiences. The study focused on designing science demonstrations with discrepant events to encourage student inquiry and exploration. The research site was the Muslim Dastageer Govt. Girls' Secondary School, with 12 participants selected through purposive sampling. Data was collected through observations, feedback from critical friends, practicum supervisors, cooperative teachers, and student reflections. Thematic analysis was done to analyze the data and to draw meaningful conclusions. The study's findings and implications emphasize the potential of demonstration through discrepant events and use of technology to enhance students' engagement, while acknowledging the limitations and the need for further research across various grade levels. The findings not only revealed the power of discrepant events in sparking interest but also provided valuable insights for educators looking to create dynamic and mesmerizing science learning environments in classroom.

Govt. College of Education, F.B. Area, Karachi

CERTIFICATE OF APPROVAL

BISMA HAYAT S/o, D/o, **HAYAT MUHAMMAD** student of B.Ed. (Hons.) Elementary has completed the Action Research entitled “**USING DISCREPANT EVENTS IN SCIENCE CLASSROOM TO ENHANCE STUDENTS’ ENGAGEMENT AT 6TH GRADE**” as the scholarly stander set in research project course guide/University of Karachi.

Signature of the Candidate: _____

Name of Candidate: **BISMA HAYAT**

Signature of the Supervisor: _____

Name of the supervisor: **PROF. TABASSUM KAUSAR**

Govt. College of Education F.B. Area, Karachi

DECLARATION

I, **BISMA HAYAT** S/o, D/o, **HAYAT MUHAMMAD** student of B.Ed. (Hons) Elementary hereby declare that no part of the thesis titled “**USING DISCREPANT EVENTS IN SCIENCE CLASSROOM TO ENHANCE STUDENTS’ ENGAGEMENT AT 6TH GRADE**” presented by me has been plagiarized from anywhere. Proper references are cited wherever necessary.

I understand that the University of Karachi reserves the right to cancel the degree if any of the above declaration is proved false before or even after award of degree.

Signature of the Candidate: _____

Name of Candidate: **BISMA HAYAT**

Seat Number: **A1-19136011**

DEDICATION:

In loving memory of my dear father, Hayat Muhammad whose unwavering love, guidance, and support continue to inspire me every day. Though he may no longer be here in person, his presence and influence will forever shape my academic journey. This research is dedicated to him, as a testament to his lasting impact on my life. I miss you, Papa." I dedicate this research to my father whose unwavering love, support, and encouragement have been the driving force behind my academic period. His sacrifices and dedication have been a constant source of inspiration and motivation throughout my journey.

My father has always instilled in me the value of education and the importance of pursuing one's passions. His support and encouragement have been instrumental in shaping the person I am today. I am forever grateful for his love, guidance, and sacrifices.

I hope that this work will make them proud and serve as a small token of my gratitude for everything they have done for me.

Thank you, Papa, for everything. This research is dedicated to you with all my love and gratitude.

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CHAPTER ONE

INTRODUCTION

1.1 Background of the study

Throughout my observation during my teaching practicum, I have found that motivating students and keeping their interest in science has been a challenge. This project employed action research to study how science demonstrations can be designed so as to provide an effective pedagogical tool to support students' productive engagement in science classroom. My prime directive as a teacher has been to find out the ways of making my students successful contributors to a rapidly changing world, and to be able to solve difficult problems.

National Science Education Standards state that "inquiry using authentic questions generated from student experiences is the central strategy for teaching science" (National Research Council, 1996, p.31), underscoring the need for educators to establish classroom environments that encourage student engagement in scientific investigations.

Constructivist theory of learning is grounded in the notion that individuals construct meaning and knowledge from their perception of prior experiences, which guide their perceptions of present experiences (Driver, Asoka, Leach, Mortimer & Scott, 1994).

The National Science Education Standards (1996) have developed new science models for teachers to integrate inquiry based instructions into their science curriculum. I want my students to develop the skills of reasoning and problem

solving that are relevant to real life and foster lifelong learning. This is in line with the recent school reforms that demand higher standards in teaching- these skills. To achieve this goal, I use collaborative pedagogical strategies in science classrooms that encourage my students to explore their environment, seek answers, and create solutions on their own.

At one point, every educator has struggled with the challenge of keeping the attention span of a classroom full of students. When students do not find subjects or problems particularly interesting, they often have difficulty staying focused and have issues with retaining important information. The arduous preparations of a teacher will remain wasted if realistic demonstrations of the scope and limitations of science are not presented. These challenges can be addressed by enhancing the interaction amidst the students and the industry. One way to overcome this hurdle and to foster a greater interest in the material being taught is to encourage students to become active contributors to their own learning experience through a strategy of science method known as “Discrepant Events”.

According to Piaget (1964), intellectual growth, or change, results from a disruption of cognitive equilibrium, established from a conflict between incoming information and what already exists in an individual's conceptual framework. Resolution of this disequilibrium results in a modification of existing knowledge schemes, leading to learning. This is conceptual change. Pedagogical approaches informed by a conceptual change model employ cognitive conflict — in which an individual's current conceptual framework is challenged by an experience - as a stimulus for learning. Such cognitive conflict is most often promoted through the use

of anomalous data - hence my decision to focus on science demonstrations that are designed around a discrepant event.

In this study, a practical example will be developed within the scope of discrepant events in science course of 6th grade students at elementary education. I am also interested in this topic as I also want to make my classroom environment as interactive as possible. Most students want to be engaged in an activity, rather than copying down definitions. While this study focuses solely on using discrepant events in science classroom.

As a student teacher I have noticed the decline in the students' interest concerning science classroom. There has been a greater emphasis placed on the necessity of technology, rather than engaging the deductive thought processes that shape science and utilize technology as a tool. Dewey (1916) believed a child is the best prepared for the demands of responsible membership within the democratic community.

Numerous studies have highlighted the positive impact of using discrepant events in science education. For instance, a study by Atasoy and Dikmenli (2016) investigated the effects of discrepant events on students' conceptual understanding and found that these events significantly improved students' conceptual understanding compared to traditional instructional methods. Similarly, in a study by Yaseen and Khan (2020), it was observed that the use of discrepant events increased students' interest and engagement in science learning, leading to higher retention of scientific concepts.

The findings from this study are expected to provide valuable insights into the effectiveness of using discrepant events as a pedagogical tool to enhance student engagement in the 6th-grade science classroom. By identifying the impact of these events on students' learning experiences and attitudes towards science, this research aims to contribute to the ongoing discourse on innovative teaching practices that promote active engagement and deeper understanding of scientific concepts among middle school students.

2.2 Statement of the problem

As a researcher of this research I have found out many of the pupils in grade 6th are lacking interest in science classroom and remain passive throughout the period. . Despite the emphasis on active learning strategies, there exists a persistent need to enhance science. Student engagement with scientific concepts, promoting deeper understanding and sustained interest in science classroom. I am hoping that by integrating creative pedagogical tool will be fruitful in capturing students' interest in science classroom. By incorporating science strategy; discrepant events will enhance their science expertise in exploration and open up a way towards lifelong learning.

As a practitioner myself, I also believe that science teachers might benefit from research that investigates how a science demonstration could be designed so as to maximize the potential of producing specific types of learning outcomes. Thus, I designed this study to focus on the effects of specific design elements within science demonstrations on students' engagement in scientific investigations. One of these key

design elements was developing the science demonstration around a discrepant event - that is, "a phenomenon that occurs in a way that seems to run contrary to initial reasoning" (Wright and Govindarajan, 1995, p. 25). As discussed in more detail in Chapter Two, anomalies are a major source of dissatisfaction (Limon, 2001) that can lead students to question their current conceptions and motivate them to look for explanations through student-generated investigations which may lead to significant learning and possibly even conceptual change. Another design element this study investigated was the role played by having students engage in explicit predictions as an integral part of the demonstration. Known in the literature as the Predict, Observe and Explain (POE) model, this strategy actively involves students in the demonstration by predicting what will occur prior to the event, observing the event, and finally attempting to explain it, verbally and/or in writing.

The problem encompasses the need to assess the efficacy of discrepant events as a pedagogical tool in cultivating students' engagement and learning experiences within the 6th-grade science classroom. Understanding the potential impact of these events on student engagement is crucial for informing educators on effective strategies to promote active learning and sustained interest in science among middle school students.

1.3 Purpose of the study

The primary purpose of this action research study is to investigate the impact of integrating discrepant events into the 6th-grade science classroom on students' engagement and conceptual understanding of scientific principles. The

study aims to evaluate the effectiveness of these events as a pedagogical tool to enhance student learning experiences and foster sustained interest in science. The purpose of the study is to measure the level of student engagement before and after the introduction of discrepant events in the science classroom through observational data and student self-reports and to assess changes in students' conceptual understanding of scientific principles by administering pre- and post-tests related to topics covered by the discrepant events. The one of the main purpose of this study is to generate recommendations for educators on the effective integration of discrepant events into the science curriculum to enhance student engagement and learning outcomes.

1.4 Research Questions

The ultimate goal of this study was to learn more about how demonstrations could be designed so as to most effectively promote students' engagement in scientific inquiry. The choice of focusing on demonstrations as a potentially powerful pedagogical tool to support student inquiry was motivated by the many research findings documenting that science demonstrations can increase student interest and motivation to learn more about a topic - and, thus, provide the catalyst for students to engage in authentic inquiry. The choice of designing each demonstration around a discrepant event came from recognizing that, as a discrepant event is likely to create cognitive conflict, it has not only the potential to further increase students' curiosity - and, thus, their willingness to engage in inquiry - but could also provide the starting point for conceptual change. Finally, recognizing that the effectiveness of science demonstrations could depend on

specific design elements, I was interested in exploring the implications of using the well-established POE (Predict-Observe-Explain) model.

The specific elements of this intervention as well as the data collection and analysis procedures described in detail in the remaining sections of the chapter were designed so as to enable me to address the following research questions using an action research approach:

- How do discrepant events impact 6th-grade students' engagement in the science classroom?
- What changes occur in students' conceptual understanding of science topics after exposure to discrepant events in the classroom?
- What role does the discrepant events play in sustaining 6th-grade students' curiosity and engagement with scientific phenomena?

1.5 Significance of the study

The significance of the action research study on using discrepant events to enhance students' engagement in a 6th-grade science classroom lies in its potential to revolutionize science education by addressing key educational challenges and leveraging innovative teaching strategies. This study has the potential to revolutionize classroom dynamics by providing educators with evidence-based insights into using discrepant events as a means to boost student engagement. Engaged students are more likely to participate actively in learning, fostering a deeper understanding of scientific concepts.

By exploring the impact of discrepant events on students' conceptual understanding, the study aims to unveil the effectiveness of these events in challenging preconceived notions and fostering a more profound comprehension of scientific principles among 6th-grade students. Inquiry and the National Science Education Standards (2000) take a strong stand supporting student's involvement in an active process of inquiry. They describe the following process skills for grade K-12:

- a) Ask questions about objects, organisms, and events in your environment,
- b) Plan and conduct a simple investigation,
- c) Communicate investigation and explanation (p. 19). Carlson

Discrepant events, which present phenomena conflicting with students' expectations, provoke curiosity and stimulate interest in science. They prompt students to question, explore, and seek explanations, thereby fostering active engagement in learning (Sampson & Grooms, 2010). Such events create cognitive dissonance, prompting students to reconcile their existing knowledge with new experiences. This process leads to deeper understanding and retention of scientific concepts (Driver & Oldham, 1986). Encountering discrepancies encourages students to think critically, analyze information, and develop problem-solving skills. They learn to evaluate evidence and construct logical explanations (Minner et al., 2010). Discrepant events align with the principles of inquiry-based learning, encouraging students to ask questions, investigate, and form hypotheses, which are essential aspects of the scientific method (NRC, 2000). Engaging in hands-on, intriguing activities fosters a positive attitude toward science, potentially

influencing students' future career choices and lifelong interest in STEM fields (Osborne et al., 2003). Action research focusing on implementing discrepant events enables educators to reflect on their teaching strategies, adapt to students' needs, and refine instructional methods, leading to continuous professional growth (Feldman, 2000).

1.6 Delimitation of the study

- i. The study specifically targets 6th-grade students, limiting the scope of the research to this particular grade level. Findings may not be directly applicable to other grade levels without further investigation.
- ii. The research focuses solely on the integration of discrepant events within the science curriculum. It does not extend to other subjects or disciplines outside the realm of science education.
- iii. The research might take place in a single or limited number of 6th-grade science classrooms, potentially restricting the generalizability of findings to broader educational contexts.
- iv. The study might use observational protocols, and interview guides tailored to the research context, which may limit comparisons with studies using different assessment tools.

1.7 Operational definitions of key terms

- I. **Discrepant events:** Demonstrations or experiments that involve unexpected or counterintuitive results, designed to challenge students' preconceptions and stimulate curiosity in science by presenting phenomena that do not align with common expectations or everyday experiences.

- II. **Engagement:** Refers to the extent of a student's involvement, interest, attention, and active participation in learning activities within the science classroom. It encompasses both behavioral and emotional aspects of a student's interaction with the subject matter.
- III. **Enhancement:** The process of improving or augmenting the quality, effectiveness, or impact of a particular aspect—in this case, the enhancement of students' engagement—within the context of science education through the incorporation of discrepant events.

CHAPTER TWO

REVIEW OF THE RELATED LITERATURE

2.1 Introduction and Overview

Developing a more nuanced understanding of the pedagogical potential of demonstrations requires us to understand the unique role of context in learning, how learners process information that does not correspond to their current understanding of the world, and the ways in which demonstrations can affect learning. Thus, to ground this study we need to look at several complementary bodies of work that shed light on these various dimensions.

Science education is a vital component of the curriculum that aims to develop students' scientific literacy, inquiry skills, and interest in science. However, many students often perceive science as boring, difficult, and irrelevant to their lives, and lose their motivation and curiosity to learn science (Osborne, Simon, & Collins, 2003). One of the challenges for science teachers is to design and implement effective instructional strategies that can capture and sustain students' attention, stimulate their thinking, and foster their engagement in scientific investigations. One of the instructional strategies that has been widely used and recommended by science educators and researchers is the use of discrepant events in science demonstrations.

2.2 What is a Discrepant Event?

A discrepant event is an unexpected or surprising phenomenon that challenges students' prior knowledge or assumptions and stimulates their inquiry (Akerson, Buck, Donnelly, Nargund-Joshi, & Wieland, 2011). Discrepant events can be used to introduce, illustrate, or reinforce science concepts and principles, as well as to generate questions, predictions,

explanations, and hypotheses among students. Discrepant events can also be used to create a sense of wonder, curiosity, and excitement in science learning, and to motivate students to explore and investigate the phenomena further.

A discrepant event is a phenomenon which has a surprising and paradoxical outcome that is not what the observer would normally expect. Consider for example, the potato which when put into two containers of identical looking liquids, remains suspended in one but sinks in the other. Or the case of the knitting needle that is pierced right through a full-blown balloon without bursting it. This kind of occurrence which runs counter to one's intuition, tends to evoke a strong feeling of "wanting to know more about how it works", and a consequent need to solve the discrepancy in one's mind. It challenges the observer to find answers to the seemingly illogical results of the event to resolve the conceptual conflict that arises (Chin Christine, Teaching and Learning, 13[1], 51-57).

2.3 Why use Discrepant Events?

In teaching a science concept or principle, it is important for the teacher to arouse the students' curiosity. The use of discrepant events is one method which can capitalize on students' curiosity as the discrepancy itself piques students' interest and compels them to seek a solution to the "mystery". When students are strongly motivated, conditions are favorable for learning. The use of discrepant events in Science has its base in Festinger's (1957) Theory of Cognitive Dissonance. According to this theory, the existence of cognitive dissonance or inconsistency creates a psychological discomfort that will motivate a person to reduce the dissonance and actively seek consonance or mental equilibrium. In the case of discrepant events, the incongruity between what is physically observed to occur

and what students think should occur upsets their intellectual equilibrium and impels them to change their existing intellectual scheme. In their search for information which will logically explain the occurrence, students are involved in the process of inquiry and problem-solving which are so essential in scientific investigations.

2.4 Situated Cognition

Situated Cognition as a theory attempts to explain how an individual's observations, interactions, and perceptions of and with an environment lead to learning (Brown et al., 1989; Seel, 2001). Brown et al. (1989) suggest that "by ignoring the situated nature of cognition, education defeats its own goal of providing useable, robust knowledge" (p. 32).

Situated Cognition considers knowledge as "inextricably a product of the activity and situations" in which it is produced (Brown et al., 1989) and regards knowledge as a set of tools that can be used in new situations and can only be completely understood through use. Using these tools involves an adoption of the culture in which they are used and results in modifications of the users perceptions of the world.

According to Brown et al. (1989), through increased use concepts continue to develop. Cognition is interdependent with the physical and social environment. At the foundation of Situated Cognition are all the internal processes that transpire as individuals interact with their environments. As such, in this study I will operationalize learning as the ability for an individual to construct meaning from one's environment.

Situated Cognition Theory asserts that individuals will actively develop their own perceptive interpretations of experiences and the objects found within them (Seel, 2001; Young, 1993).

The intervention at the core of this study involves students in demonstrations and investigations that provide opportunities for revision and construction of a specific concept. This is consistent with Brown et al.'s (1989) belief that "People who use tools actively rather than just acquire them, by contrast, build an increasingly rich implicit understanding of the world in which they use the tools and of the tools themselves" (p. 33) and that concepts continually develop as "new situations, negotiations, and activities inevitably recast it in a new, more densely textured form" (p. 33).

Young (1993) concludes that from a situated cognition perspective, it is necessary for teachers to draw student's attention to the meaningful features of a situation or problem. Baddock and Bucat (2008) agree and conclude that along with being explicit, teachers should be repetitive and dramatic, so that students can distinguish the important from the unimportant information in the demonstration. Without this, the authors believe that observations, existing knowledge and details of the demonstration might establish a cognitive overload for students.

2.5 Conceptual Change Theory

Conceptual Change Theory posits that learning occurs "against the background of the learner's current concepts" (Posner et al, 1982, p. 212) used to organize new experiences. Essentially, conceptual change involves the reorganization of an "interrelated system of beliefs" (Vosniadou et al, 2001, p. 394).

Introduced by Piaget (1964), accommodation is a process of conceptual restructuring occurring through modification of existing knowledge. Essentially, it occurs when students use existing concepts to understand new phenomena. On the other hand, assimilation is the incorporation of new information into existing conceptions.

Piaget (1964) explains that learning can be stimulated when cognitive dissonance occurs, or when an observer experiences a situation that contradicts expectation. The resulting state of perplexity and doubt, called disequilibrium, plays a significant role by stimulating curiosity and engaging the learner. The observer is compelled to seek information that can explain the occurrence, due to the mental discomfort of two simultaneously existing competing concepts.

Discrepant events have been identified as a valuable means of causing cognitive conflict in students, as they represent a type of anomalous experience that compels students to focus on prior conceptions, a necessary step for conceptual change (Pintrich, Marx & Boyle, 1993). Even if no conceptual change results, anomalous data support at least the initial steps towards conceptual change (Limon, 2001) by promoting reflection and initiating awareness and self-recognition of ideas and assumptions (Limon & Carretero, 1997).

The discrepant event demonstrations were designed so that students would encounter dissatisfaction during observation, explore plausible explanations through their own investigations, and finally select a fruitful explanation through experience and discussion. It was the intention of the study to operationalize cognitive conflict in students through the discrepant event demonstrations.

2.6 Inquiry-Based Reform

A significant amount of research, as reported in the National Science Education Standards (National Research Council, 1996), supports an inquiry-based approach to science instruction. As the foundation for my study, it was imperative to learn more about

how demonstrations can help support inquiry experiences. The National Science Education Standards define "full inquiry" as a process in which students:

1. Pose a productive question;
2. Design an investigation directed toward answering that question;
3. Carry out the investigation, gathering the applicable data in the process;
4. Interpret and document their findings;
5. Publish or present their findings in an open forum.

The literature contains extensive empirical support demonstrating that inquiry conducted in the classroom leads to meaningful student learning (Anderson, Reder, & Simon, 1996). The literature also identifies certain features as characterizing quality scientific investigation (Chinn & Malhotra, 2002). Utilizing an inquiry-based approach, the intervention at the core of this study incorporates these features and strategies in a number of ways.

Palincsar (1989) suggests that, when Brown et al. (1989) speak about the interdependent nature of activity, concept, and culture, they are advocating bridge-building, in the sense that an individual can acquire the awareness and ability to translate and apply knowledge learned in one situation to one that is distinctly novel. This capacity to generalize will be examined in student- drawn conclusions. In today's technological age, it is critical for students to gain an appreciation for locating information, and knowing how to apply it. For knowledge to be considered a tool (Brown et al., 1989) it must be amenable to various situations (Young, 1993), a strength that may become apparent in journal entries and interviews.

2.7 Research on Demonstrations

The science demonstration is a widely used tool in science curricula at all grade levels (Glasson, 1989). The design of a traditional science demonstration typically involves a teacher or student performing an activity while the rest of the class observes (Milne & Otieno, 2007). Research shows that demonstrations are essentially used by educators to achieve three primary goals: the first is to illustrate or legitimize a concept, the second is to promote student comprehension, and the third to increase student engagement in the classroom (Beall, 1996; Candela, 1998; Manaf & Subramaniam, 2004; Morgan, Barroso, & Simpson, 2007). Research has shown that demonstrations in the chemistry classroom stretch students "cognitive spectrum" through their ability to capture attention (Buncick, Betts, & Horgan, 2001), spark curiosity (Shepardson, Moje, & Kennard-McClelland, 1994), generate interest (Callan, Crouch, Fagen, & Mazur, 2004), and develop higher level thinking skills, promoting "functional understanding of various concepts" (Meyer et al., 2003, p. 432). Science demonstrations have been shown to be effective in "capturing attention, igniting curiosity and promoting functional understanding of various concepts" (Manaf & Subramaniam, 2004, P-1).

Others argue that science demonstrations promote active learning environments (Manaf & Subramaniam, 2004) and can be an effective strategy for learning to occur when students are actively involved, such as predicting and discussing observations (Furtak, 2009; Morgan et al, 2007). Beasley (1982) reported by Fagen (2003) found that demonstrations increase levels of student attention and task involvement leading to increased student learning and understanding of intended concepts. Both instructors and students have claimed to learn from demonstrations (Freier, 1981; Hilton, 1981; Morgan et al, 2007). However, critics of traditional science demonstrations maintain they are

inadequate methods of instruction where students, as passive observers, have difficulty grasping underlying concepts (Callan et al, 2004; Lynch & Zenchak, 2002; Roth, McRobbie, Lucas & Boutonne, 1997) and are "less motivated to solve problems independently" (Glasson, 1989, p. 129). These arguments are grounded in the belief that demonstrations offer nothing more than momentary, attention-getting displays and, rather than engaging students in true learning experiences, simply capture fleeting attention due to their visual appeal. Thompson (1989) acknowledges that demonstrations are oftentimes misused by science teachers who too often use these as "fun activities", rather than illustrations of concepts. Criticism has also labeled traditional science demonstrations as inquiry stifling, charging they do not allow students to explore "what if questions (Morgan et al, 2007).

Roth et al. (1997) report that one reason students fail to learn from teacher demonstrations designed with a traditional transmission approach to teaching is that they are not provided opportunity to test their explanations. Shepardson et al. (1994) also conclude that demonstrations should not be viewed as "an end in themselves, but should lead to the testing of children's ideas" (p. 255). Therefore, in the intervention at the core of this study, student-designed inquiry experiences allow students to design and implement an investigation in an effort to explain an observed discrepant event phenomenon. Studies have investigated the effects of demonstrations as a means to increase student attention and task involvement (Beasley, 1982); foster inclusivity (Buncick et al, 2001); develop students' critical thinking (Meyer et al, 2003); support students' abilities to write predictions, make observations and develop explanations (Shepardson et al, 1994; Furtak, 2009); and improve students' test scores. Milne and Otieno (2007) claim, however, that

much of the research done on demonstrations examines learners understanding of specific science content, rather than examining the structure or purpose of the demonstration as the focus.

There is literature regarding variation in the strategies that educators can employ during classroom demonstrations. They can be used as an introduction to a topic, as a wrap up, and as a tool throughout the class discussion of the topic. Demonstrations have also been studied as a means to review, reinforce, and relate concepts. Morgan et al. (2007) found that different types of learning outcomes are achieved in each of these situations.

In a Predict-Observe-Explain (POE) demonstration students are introduced to an experimental scenario and are asked to predict what they think will happen. The teacher next conducts the experiment while the students observe and record their observations. Finally, students are asked to discuss their initial predictions, how they differ from the actual results, and to explain their reasoning behind their predictions. The central tenet of this design lies in its strategy to involve students actively in the demonstration experience by predicting what will occur prior to the event, observing the event, and finally attempting to explain it, verbally, in writing and through discourse.

The Predict-Observe-Explain (POE) strategy has been shown by research to be an effective instructional tool that engages students and strengthens their understanding of science concepts through the use of demonstrations (Furtak, 2009). The POE strategy provides opportunities for students to demonstrate what they know and for misconceptions to be identified.

2.8 Research on Discrepant Events

Consistent with a conceptual change paradigm, McDermott (2001) found that for students to identify and correct misconceptions, student ideas need to be explicitly elicited, students must be explicitly confronted with the errors involved in their thoughts, and students must be provided the opportunity to address these ideas and errors (Crouch et al, 2004).

Wright and Govindarajan (1995) define a discrepant event as "a phenomenon that occurs in a way that seems to run contrary to initial reasoning" (p. 25). Research has shown the most effective implementation of discrepant events involves teachers who neither confirm nor deny students explanations, but rather guide the student towards, and through, fruitful investigations to evaluate these explanations (Candela, 1998; O'Brien, 1991).

Based on these considerations, each discrepant event used in this study was chosen to reflect the following characteristics:

1. It was designed to engage and motivate the observer to want to know more, investigate more, or at the very least, to want to continue to observe.
2. An explanation for the observed phenomena could not occur without further investigation.
3. The experience would be categorized as surprising, startling, paradoxical, amazing, puzzling, bewildering, confounding, strange, unusual, unreal, bizarre, shocking or attention-grabbing.

2.9 Recent Researches on Discrepant Events:

For many years, researchers have been interested in studying the cognitive aspects of science education, particularly how students achieve conceptual change. To this end, they have developed several teaching strategies that are easily applicable. One of these

strategies is called discrepant events, which are powerful tools that can stimulate interest, motivate students to challenge their covert science misconceptions, and promote higher-order thinking skills. The key point is that directly challenging students' naive ideas will lead to more quality science learning going on in the classroom. (Wilson J. González-Espada et al, 2010).

Discrepant events are a powerful way to activate thinking and challenge students' misconceptions. A discrepant teaching event is similar to a discrepant science event in that it vividly portrays what is often an abstract construct or concept and has an unexpected outcome. The unexpected outcome creates what Piaget (1971) refers to as disequilibrium, thereby uncovering students' naïve conceptions and tacit beliefs about the concept being studied. This article defines what a discrepant teaching event is and compares and contrasts discrepant science events and discrepant teaching events. The article also provides examples of discrepant teaching events that are useful in mathematics and social studies. Finally, the article concludes with a discussion of the utilization of an "inquiry stance" to teaching as a way to address students' misconceptions of discipline-specific concepts. (Longfield, Judith. 2009). The study revealed that incorporating discrepant events in a 10th grade biology classroom positively influenced both student academic achievement and engagement. Students who took part in these events demonstrated higher test scores and greater classroom involvement compared to those who did not participate. Additionally, the research indicated that the use of discrepant events positively affected the teacher's attitude towards teaching, leading to increased enthusiasm for the subject matter. (Rouwenhorst, L. K. 2017)

The study by Doğan, Ş., AKTER, B., & ÇOBAN, G. (2023) aimed to explore the impact of incorporating inquiry practices involving discrepant events on the conceptual understanding, inquiry skills, and perspectives on scientific knowledge among pre-service science teachers. The findings revealed no statistically significant difference in prospective teachers' views on scientific knowledge and inquiry skills. However, a noteworthy difference was observed in their conceptual changes. The study's outcomes offer implications for understanding the results.

CHAPTER THREE

RESEARCH METHODOLOGY

3.1 Research Design/Strategy

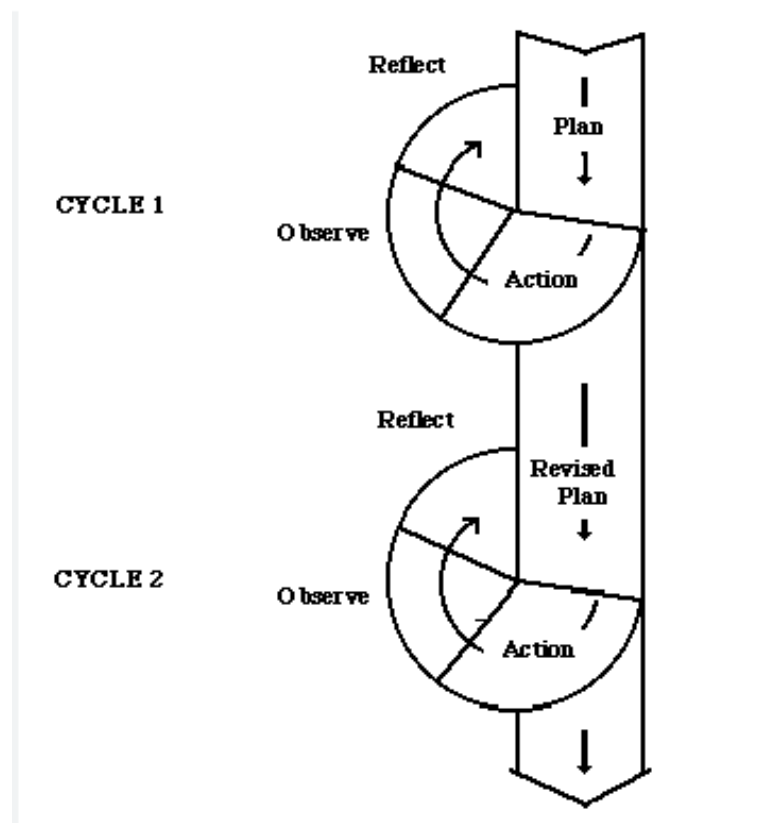
This chapter discusses the research methods that the researcher will use to conduct the study. It covers aspects such as the research design study population and Research instruments. This Study was made by the means of qualitative methods of research. The researcher attempted to get the answer to the mentioned problem and to justify and satisfy the objectives of the study. Qualitative research is generally more explorative a type of research that is dependent on the collection of verbal behavior or observational data that can be interpreted in a subjective manner. Action research a type of qualitative research emphasizes collaboration and practical problem solving within a specific context according to (Kemmis and Mc.Taggart 1988) action research involves a cyclic process of planning acting observing and reflecting allowing for continuous refinement and improvement this approach aligns with the notion that research should be a dynamic participatory process rather than a detached observation (Carr and Kem in 1986).

For the description of this study, I decided to opt an action research. Action research: It is an involvement of teachers identifying a topic or problem study, collecting and analyzing information to solve or understand a teaching problem or to understand the aspects of their practices (Educational research 2003, P.262). Action research is designed to bridge the gap between research and practice (Somekh 1995: 340).

With the action research approach, I was able to assess and evaluate the responses of students. Classroom action research was used in accordance with the purpose of this study. According to Kemmis & Mc Taggart (1992), action research is to do plan, act, observe and

reflect more carefully and systematically and more rigorously than one usually does in everyday life. According to him, action research is consist of four phases:

- 1- **Plan:** Once the problem is identified the researcher plan the strategies and actions to be undertaken to address the identified issue or area of improvement.
- 2- **Act:** In this step the planned action are implemented in the real setting, this involve executing the planned strategies to bring change or improvement.
- 3- **Observe:** In this step the researcher gather data during the implementation and analyze the effects of the action, this step involves collection of data
- 4- **Reflect:** After gathering and observing data of the actions the researcher engage in reflection. This involves analyzing the data to know about what worked well and what could be improved.



3.2 Research site

The Muslim Dastageer Govt. Girls' secondary school was a research site of my study in academic year 2022-2023.

3.3 Participants of study

The participants of my research study were the students of 6th grade science class, one of my critical friend & the cooperative teacher who observed my lessons. The sample population include 12 girls. The type of sampling used for this research was a purposive sampling. The participants shared the approximately same age and ability level.

3.4 Research Instruments

I have used several research instrument for my research study as I wanted to collect qualitative data. Throughout the intervention I collected the following complementary data in order to address my research questions.

3.4.1 Classroom Transcripts

Each lesson, was video recorded to capture students' engagement and reactions to the various demonstration designs. This included the introduction of the demonstration, the class discussion following it, and the inquiry question.

3.4.2 Students' Reflective Journals

Paper sheets were available for each student to write in entries in their reflective journals. This work was saved serving as artifacts to be used for later analysis and reference (See Appendix A.1). It is important to note that as well as a valuable data source, student responses written in reflective journals were an integral part of the inquiry experience. The reflective

journals acted as opportunities for students to build their learning of investigative techniques, and of a particular concept, as well as opportunities for students to develop the investigation itself in a progressive, systematic way. At the end of the lesson, they helped students to reflect on how the lesson affected their understanding and learning of science and the particular concept at hand. For the teacher, these reflective journals offered formative assessment tools throughout the process and summative tools at the end of each unit, as student reflections provided a comprehensive collection of experiences.

3.4.3 Other Student Work

I collected all the other written work provided by the students in class, including worksheets.

3.4.4 Teachers' Reflective Dairy

At the end of each day of the intervention lessons, I wrote personal notes in a journal as a way to record a "teacher's perspective" on these experiences. These notes included important decisions made in the planning and/or execution of these lessons, my observations about students' reactions to the experiences, suggestions for future presentations, and anything else that seemed relevant to record, to ensure a systematic process.

3.4.5 Observer's Checklist

In each unit, the initial demonstrations/lecture was also observed by a co-operative teacher and one of my critical friend. Student's non-verbal body movements are telling ways to identify engagement and this observer helped recognize these indicators during the lesson, when I was not able to do so because my attention needed to be focused on doing the demonstration or lecture. Throughout the demonstration or lecture, the observer marked on the checklist any

student who showed evidence of engagement/interest as to provide a qualitative and easily comparable measure of students' responses to the different instructional designs employed (See Appendix A.2). The observer's presence at each of the demonstrations/lectures established a consistency that provided a richer comparison.

3.4.6 Transcripts of Final Class Reflection

The final class reflection that took place in each of the lesson was audio-recorded and fully transcribed by me.

3.4.7 Transcripts of Semi-Structured Interviews

Co-operative teacher of grade 6th that I taught were selected and invited to participate in an interview, which took place after class, following the final class reflection. The co-operative teacher was selected because I thought her final class reflection comments was particularly thoughtful and insightful. This interview was fully transcribed (see Appendix A.3)

3.5 Procedure of Data Collection

For my action research the data collection process involves employing several tools to gather comprehensive insights for action cycle 1 & action cycle 2. The primary tools of action cycle 1 were the supervisor observation sheet, a checklist designed for the cooperative teacher as well as for critical friend and semi-structured interview for students. But for action cycle 2, I only used cooperative teacher's, critical friend's and students' reflections.

The supervisor observation sheet provides a systematic way for the supervisor to keep track of how teaching and learning are going. Moreover, the checklist designed for the cooperative teacher and critical friend facilitates a thorough evaluation of essential elements

connected to cooperation and assistance within the research setting. Furthermore, conducting semi-structured interviews with students offers a deeper insight into their viewpoints, experiences, and thoughts about the demonstrations being used.

3.6 Procedure of Data Analysis

The data was analyzed through thematic analysis in both action cycle 1 and action cycle 2. Thematic analysis is a technique for assessing qualitative data which involves looking through a data collection to find, examine and report recurring themes (Braun and Clarke, 2006). Thematic analysis is a way of analyzing qualitative data where researchers go through the data carefully, looking for common patterns or themes in the information they've collected. These themes represent recurring ideas or concepts that help understand what the data is trying to convey. It's not just about identifying patterns; it also involves the researcher reflecting on their own thoughts and feelings, acknowledging that their own experiences and perspectives influence how they interpret the data. This method allows researchers to organize and make sense of the data by identifying key themes, which helps in drawing meaningful conclusions or understanding the main ideas present in the information collected.

3.7 Ethical Consideration

Before the visit researcher will properly take permission from the management of teacher education institution and will follow the rules and regulation of sample institution. The researcher can conduct the research with maximum attention to maintain the objectivity of research process all the participants will be aware about the nature and purpose of this research and all the personal information related to the participants will be kept confidential.

CHAPTER FOUR

DATA ANALYSIS AND REFLECTIONS

The findings from this study are expected to provide valuable insights into the effectiveness of using discrepant events as a pedagogical tool to enhance student engagement in the 6th-grade science classroom. Data was gathered during the implementation of action cycles 1 & 2.

4.1 Negotiation

First of all I took consent to the principal for this action research study and also her permission for some special classes (see Appendix E). I also selected one of my critical friend & a co-operative teacher for mentoring also.

4.2 Reconnaissance

After negotiation, I started my fieldwork and start observing the students in science classroom. I observed that students were not taking interest in science subject and they got bored in just few minutes. The student teacher who was my colleague at the practicum session, using different A.V. aids for students but still these A.V. aids were not enough to grab the students' attention and interest. Following are the findings of reconnaissance:

Teacher used A.V. aids which was not enough for grabbing students' engagement in science classroom. Her focus is just to complete the lesson on time so she didn't give an opportunity to a student for deep thinking and questioning. Even teacher was using visual aids but student still feel passive in science class. She was not using any digital tool to make connections of students to real world of science. Also she avoid using new strategies in science

classroom as it takes time. Students were not able to go through deeper understanding of the concepts in science classrooms.

4.3 Action cycle 1

After analysis of my observation, I started action cycle 1 for the improvement of students' engagement in science classroom (see Appendix D.1).

I introduced the new strategy in science class which is discrepant events and provide students an opportunity to POE (predict what happen, observe what happening, and explain what happened) during the demonstrations.

4.3.1 Observations of Action Cycle 1

During action cycle 1 researcher noted the following observations:

1. I observed that discrepant events helped in grabbing the students' attention in science classroom
2. The co-operative teacher easily observed researcher through checklist and she gave the so positive response towards the use of discrepant events in science classroom.
3. The critical friend was seem so helpful in this research project her responses helped researcher in noting the strengths and weaknesses of the approach.
4. The practicum supervisor was observing researchers teaching skills but her responses also add as a beneficial for researcher for the improvement in students' engagement.
5. Students gave their responses in a positive way that they love to learn through this technique as they were able to understand the concepts of science.

4.3.2 Analysis of My Observation

The student's responses indicate a strong positive reaction between surprise elements in science lessons and heightened engagement, curiosity, and an eagerness. I observed that in very first class when I introduced them a discrepant event related to the topic about the weight of air, first they denied to accept that air has mass but when I give evidence of this through discrepant event, they got confused and start discussion with me that how it happens. It showed the student's involvement in the classroom. It was a very positive reaction of students. But still they feel passive throughout the event I performed in front of them. I observed that students just agree with the event as teacher tell them to accept, they did not ask any questions related to the event or the topic.

4.3.3 Analysis of Students' Reflections:

The effectiveness of using discrepant events help in enhancing student interest and learning in science. One of my students said, "Teacher this is very good I really do not notice this before". They said that they would like to study through this technique daily as they had got to know new things through my lesson. They were so relaxed and engaged in the lesson as they did not have a burden of doing homework or classwork so they just listen to me and showed the highest involvement in science classroom. They also said that they never learned science through this technique before. They got shocked when I showed them a mass of air, as they thought that air has no mass. Overall, they enjoyed the science class.

4.3.4 Analysis of Critical Friend's Reflection

The feedback from critical friend is mostly positive, showing that the events aligned well with goals and were implemented effectively. She noted that the events had a positive impact on student engagement and the teacher's practice, which is great news. However, there are some

areas to work on, like ensuring the events suit all students' levels and linking them more closely with the curriculum. The feedback also suggests that we might need to dig deeper into analyzing and interpreting the outcomes. It's clear that having a critical friend in this project has been incredibly helpful in guiding me toward improvement and highlighting the strengths and weaknesses of my approach.

4.3.5 Analysis of Reflections of Co-operative Teacher:

I analyzed the responses of the co-operative teacher as her responses indicate that she had a positive and collaborative experience with the discrepant events and the action research project. She agreed or strongly agreed with all the statements in the observation checklist, which suggests that she was satisfied and impressed with the design, implementation, documentation, analysis, review, and impact of the discrepant events. The co-operative teacher's responses also provide rich and detailed information about the students' reactions and responses, the effects of the discrepant events on the students' engagement and learning, the effects of the discrepant events on my teaching practice and pedagogy. She highlighted the strengths and weaknesses, the challenges and opportunities, and the suggestions and recommendations for the discrepant event.

4.3.6 Analysis of practicum supervisor's feedback

This research project was done on the practicum session so there was one of my practicum supervisor who observed my teaching skills through a rubric. The rubric consists of several points like appearance, voice, interactionist skills, students' engagement, use of A.V aids etc. I analyze the responses of my practicum supervisor by the rubric which showed that I have a good content knowledge and voice was clear to all students. I skillfully deal with the A.V. aids throughout the lesson and students were taking interest in the classroom. Responses also indicate

that students were not ask any question to the teacher as they were just listen to the teacher and showed interest.

4.3.7 Conclusion of Action Cycle 1

After completion of action cycle 1, I realized that students were taking interest in science classroom but still not having inquiry skills about the events. They did not ask questions about the event. I need another action cycle for the better engagement of students in science classroom.

4.4 Action Cycle 2

After analysis of cycle 1 which showed some weaknesses in students' interaction in science classroom, I started action cycle 2 for the improvement of students' engagement in science classroom (see Appendix D.2).

I introduced the same strategy of discrepant events along with the integration of technology i.e. laptop. I used laptop in my lessons to show some real world experiences to the students. First I displayed the discrepant event to the students on laptop and in middle of the video I paused the video and ask students to predict what will going to happen then make sure that students ask questions about the event. I also explained them the concepts showing in the video.

4.4.1 Observations of Action Cycle 2

During action cycle 2 researcher noted the following observations:

1. I observed that use of discrepant event along with the technology in the lesson plan helped in grabbing the high engagement of students in science classroom.

2. They like to watch a video on laptop and also asked questions about the events happening in the video to the teacher.
3. The co-operative teacher also observed that students got engaged in the video and try to explore and understand the concepts which teacher trying to taught them.
4. The critical friend who was also there in class also got shocked having seeing the high participation of students in science classroom.
5. The practicum supervisor who observes the researcher's/ teacher's teaching skills also commented that teacher maintain classroom discipline so well that students engaged in the lesson. Use of technology as an A.V. aid played an important role in catching the students' attention and teacher was giving an opportunity to every students to ask questions and explore the concepts of science.

4.4.2 Analysis of My Observations

In action cycle 2 the students show a strong positive reaction between surprise elements in science lessons and heightened engagement, curiosity, and an eagerness to explore and understand the scientific concepts. I observed that when I used discrepant events on laptop in my lesson it showed the high student's involvement in the classroom. It was a very positive reaction from students. They asked questions related to the events. I observed that now students not just agree with the event they also wanted to know more about the scientific understanding as they were experiencing the real world through technology. There were some students who asked questions related to the event or the topic. Overall I observed that students love to learn through discrepant event through technology as they had an opportunity to see a scientific world.

4.4.3 Analysis of Students' Reflections

The effectiveness of using discrepant events along with the technology help in enhancing student interest and learning in science. When I open laptop during lesson one of my students said, “Wow teacher, are we learn today through laptop today?” I replied her, “Yes” then she said, “It’s going to be very interesting”. They said that they would like to study through this technique daily as they got to know new things through this lesson. The level of happiness and excitement when I displayed a video of selected discrepant event and asked them to predict what happened next in the video showed the highest involvement of student in science classroom. They also said that they never learned science through this technique before. They also demanded me to show some more evidences of that event which showed the eagerness to explore the scientific understandings in students. Overall, they enjoyed the science class and learned how to explore more through technology.

4.4.4 Analysis of Critical Friend’s Reflection

The feedback from critical friend in cycle 2 is mostly positive, showing that the events aligned well with goals and were implemented effectively. She noted that the events had a positive impact on student engagement and the teacher’s practice, which is great news. She added that use of discrepant events along with technology had a great experience for students and this technique was suitable for all students as all are comfortable with technology. In feedback she mentioned that use of technology helped to dig deeper into analyzing and interpreting the outcomes. It’s clear that having a critical friend in this project has been incredibly helpful in guiding me toward improvement and highlighting the strengths and weaknesses of my approach. She appreciated my efforts throughout the project which also helped me in analyzing the needs of students. She also mentioned that I designed the lesson plan

which suited to all students in classroom and make sure to select videos of discrepant events for the lesson plan which was easy to understand to all students.

4.4.5 Analysis of Co-operative Teacher's Reflection

I analyzed the responses of the co-operative teacher as again her responses indicate that she had a positive and collaborative experience with the discrepant events and the action research project. Mostly she strongly agreed with all the statements in the observation checklist, which suggests that she was satisfied and impressed with the design, implementation, documentation, analysis, review, and impact of the discrepant events. The co-operative teacher's responses also provide rich and detailed information about the students' reactions and responses, the effects of the discrepant events on the students' engagement and learning, the effects of the discrepant events on my teaching practice and pedagogy. She highlighted the good points of using discrepant events through technology has a great impact on students' engagement in the science classroom. She noted that students were not only taking active part in classroom but also asking questions which showed their curiosity and eagerness to explore the scientific concepts. Overall the responses of co-operative teacher showed that use of discrepant events along with the technology had a huge impact on students' engagement and interest in science classroom.

4.4.6 Analysis of Practicum Supervisor's Feedback

As I stated before that this research project was done on the practicum session so there was one of my practicum supervisor who observed my teaching skills through a rubric. The rubric consists of several points like appearance, voice, interactionist skills, students' engagement, use of A.V aids etc. I analyze the responses of my practicum supervisor by the rubric which showed that I have a good content knowledge and voice was clear to all students. I

skillfully deal with the A.V. aids throughout the lesson and students were taking interest in the classroom. In this action cycle I used technology to show the video of discrepant events to the students so my supervisor added in her comments that use of discrepant through technology had a great impact on students' engagement. She noted that teacher had a good interaction with students and she also asking questions to the students during lesson. Responses also indicate that students were ask several questions to the teacher about the event which showed their interest in science classroom.

4.4.7 Conclusion of Action Cycle 2

After completion of action cycle 2, I realized that students were taking interest in science classroom and also ask questions during lesson. They do ask questions about the events and had a great engagement in science classroom. The use of discrepant event along with the technology helped in grabbing the attention, interest, and high engagement in science classroom.

CHAPTER FIVE

CONCLUSION AND IMPLICATIONS

I begin this chapter by presenting a discussion of the study, concluding the results, discussed the implication of the study and also give the limitations of this study.

5.1 Discussion

This study was developed using action research methodology to explore how science demonstrations can be designed to most effectively promote student engagement in scientific inquiry. A key component to the design of this study was the use of discrepant events in each of the demonstrations involved, intended to motivate students to question the observed difference and promote further exploration through self-generated inquiry. The study's focus on investigating the effectiveness of using discrepant events to enhance student engagement in a 6th-grade science classroom is highlighted. The research design involved action cycles, which included the use of technology and student-centered learning strategies. This approach allowed for the systematic implementation and evaluation of discrepant events within the classroom setting.

The findings of the study suggest that the implementation of discrepant events led to increased student engagement, participation, and interest in science lessons. As also mentioned in the previous study that these captivating demonstrations, characterized by unexpected outcomes or contradictions, act as cognitive stimulants, piquing students' interest and drawing them into the scientific inquiry process (Tobias & Raphael, 1996). This indicates that the use of these events had a positive impact on the students' overall involvement and enthusiasm for learning scientific concepts. Empirical studies by Gabel and Sherwood (1980) showcase that the element of surprise inherent in these events triggers curiosity, prompting students to actively

participate and inquire about the observed phenomena. Additionally, the study acknowledges the role of technology and student-centered learning approaches in contributing to the positive outcomes observed. This suggests that the combination of discrepant events with technology enhanced the overall effectiveness of the instructional techniques employed in the study. The interactive nature of these events, combined with their unexpected nature, triggers a sense of wonder and excitement, effectively boosting students' engagement and participation in classroom discussions and activities (Beeth, 1998; Tobias & Raphael, 1996).

A rich set of complementary data was collected from a variety of sources including class transcripts, participants' reflections, critical friend's reflections, co-operative teacher's reflections and a teacher's log. Analysis of this data disclosed emerging themes and findings that helped answer each of the research questions.

Findings from this study revealed that instructional technique; discrepant events investigated influenced student engagement and learning outcomes in valuable yet distinct ways. First of all, the manner in which demonstrations are presented seemed to show evidence of student engagement in different ways. Regarding changes in students' conceptual understanding of science topics, exposure to discrepant events correlates with substantial positive shifts. Osborne and Freyberg (1985) stress the significance of hands-on experiences and real-world connections in science education. Discrepant events provide tangible, experiential learning opportunities that bridge abstract deepens as they witness these phenomena firsthand, establishing connections between theoretical knowledge and practical applications (Osborne & Freyberg, 1985).

The use of discrepant events as the content of a demonstration was a critical aspect of the intervention at the core of this study. Discrepant events not only motivated students to pursue a deeper understanding of the phenomenon but also encourage them to learn new things. Research by Renner and Lawson (1973) also emphasizes that the unexpected and surprising nature of these events captures students' attention and sustains their interest in scientific inquiry. Pictures in video, which may have also represented a discrepant event had a great impact on students' engagement and understandings. Student interest generated by discrepant event demonstrations also led to continued interest and self-directed engagement with the demonstration or the concept, such as replicating the demonstration at home, or even talking about the phenomenon at home or with the teacher after class.

As research has demonstrated that use of discrepant events along with the technology helped teachers in capturing the attention of students in science classrooms. The events trigger students' inherent curiosity, promoting autonomous engagement and a genuine interest in learning about scientific phenomena without the need for external rewards or coercion (Deci & Ryan, 1985). The final analysis has showed that to get the engagement of students in science classroom, teachers had to make efforts through using different strategies as I mentioned earlier in the study.

5.2 Conclusion

I organized this study to figure out the best way to use surprising demonstrations or discrepant events to help students get more involved in science experiments. I tried different ways of showing these surprising events and looked at how they affected students' interest in doing their own investigations. During this study, I found out that using

demonstrations in lessons can have a bigger impact on how interested students are and how they approach their own experiments than I initially thought.

Educational psychologists like Jean Piaget and Lev Vygotsky emphasized the importance of cognitive conflict and social interaction in the learning process. Piaget's theory of cognitive development highlights the role of disequilibrium, where encountering discrepancies between existing knowledge and new information stimulates intellectual growth. The use of discrepant events aligns with this theory, inducing cognitive dissonance that prompts students to actively reconcile their understanding, leading to cognitive advancement.

Considering all of the facts, the study's evidence-based insights into the effectiveness of using discrepant events to enhance student engagement in a 6th-grade science classroom are summarized. The findings of the study suggest that the integration of discrepant events, along with technology and student-centered learning strategies, can lead to increased student engagement, participation, and interest in science lessons. This indicates that the combination of these elements positively influenced the students' overall involvement and enthusiasm for learning scientific concepts.

Another Research conducted by Tobias and Raphael (1996) also highlights the effectiveness of discrepant events in science education as well. Their study emphasizes how these events, characterized by unexpected outcomes or contradictions, provoke curiosity and stimulate students' intellectual engagement. This aligns with the findings of the current study, where the surprise element of these demonstrations heightened students' interest and propelled them towards active participation and inquiry.

Furthermore, the final observations of this study emphasizes the importance of creating a classroom environment that encourages student engagement in scientific investigations. This highlights the significance of fostering an interactive and participatory learning atmosphere where students are actively involved in exploring and understanding scientific phenomena. The study's results underline the value of promoting student engagement as a key factor in enhancing the learning experience and generating interest in science among 6th-grade students.

The important thing I learned from this study is that demonstrations aren't just about getting students interested - they can actually shape how students learn and do their own investigations. When we understand all the different parts of a demonstration, how it connects to the lesson, and how students react to it, we can use it better to get students excited about learning. Overall, what I found is that demonstrations are a powerful way to help students learn by doing experiments and exploring things on their own.

Taking everything into account the study were reinforces the findings regarding the positive impact of integrating discrepant events, technology, and student-centered learning strategies on student engagement and interest in science. It emphasizes the potential of these elements to create an environment that fosters active participation and curiosity, ultimately contributing to a more engaging and effective science education experience for students.

5.3 Implications

Discussing the implications, the study's findings are discussed in terms of their potential impact on science education. The section highlights the implications of the study for educators, students, and the broader field of science education.

Firstly, the study's findings suggest that educators can control the insights provided by the study to innovate their teaching strategies and create more engaging and interactive science lessons. This indicates that the integration of discrepant events, and technology can be an effective approach to enhance student engagement and interest in science.

Secondly, the study's positive shift in students' attitudes toward science has implications for their future career choices and lifelong interest in STEM fields. This highlights the potential of the study's findings to contribute to the development of a more scientifically literate and engaged society.

Thirdly, the integration of discrepant events into the science curriculum can lead to a deeper understanding of scientific concepts among 6th-grade students. This suggests that the use of these events can be an effective pedagogical tool to promote a more profound comprehension of scientific principles among middle school students.

Finally, the study's findings emphasize the need for continuous professional growth for educators. This highlights the importance of ongoing professional development to ensure that educators are equipped with the necessary skills and knowledge to effectively integrate innovative teaching strategies into their classrooms.

In the nutshell, the implications underlines the potential impact of the study's findings on science education. It highlights the potential of the study's insights to contribute to the development of more engaging and effective science education practices, ultimately benefiting both educators and students.

5.4 Limitations

There are some limitations inherent with the design of this study that need to be taken into consideration. These limitations are important to consider in order to understand the scope and applicability of the study's results.

Firstly, the study's sample size was relatively small, and the research was conducted in a single school. This suggests that the findings may not be broadly generalizable to other educational settings. The limited scope of the study's sample and setting indicates that caution should be exercised when applying the results to a wider population.

Another constraint was noticed in the students' reflections, where it seemed that certain conclusions might have been copied and written the same way by partners, despite clear instructions that each person was supposed to write their own conclusions.

These findings shouldn't be seen as the ultimate way for all classrooms or as the perfect approach for every teacher. They were influenced by how I run my own class and teach. Based on what I learned, these methods might suit teachers who are open to taking risks and letting students take more control, but not everyone might be comfortable with that. Every classroom has its own challenges. Sometimes it's hard to find the right demonstration for a specific concept or to have enough time for both the demonstration and the follow-up investigation.

Additionally, the study's focus on 6th-grade students may restrict its applicability to other grade levels. The specific age group targeted in the study may not fully represent the experiences and responses of students in different age groups, highlighting the need for further research across various grade levels.

5.4.1 Further Research/ Research Gap

Given these limitations, the study suggests that future research should address these constraints and explore the effectiveness of using discrepant events in different contexts and with different age groups. This recommendation underscores the importance of conducting further research to expand the understanding of how discrepant events impact student engagement and interest in science across diverse educational settings and student populations.

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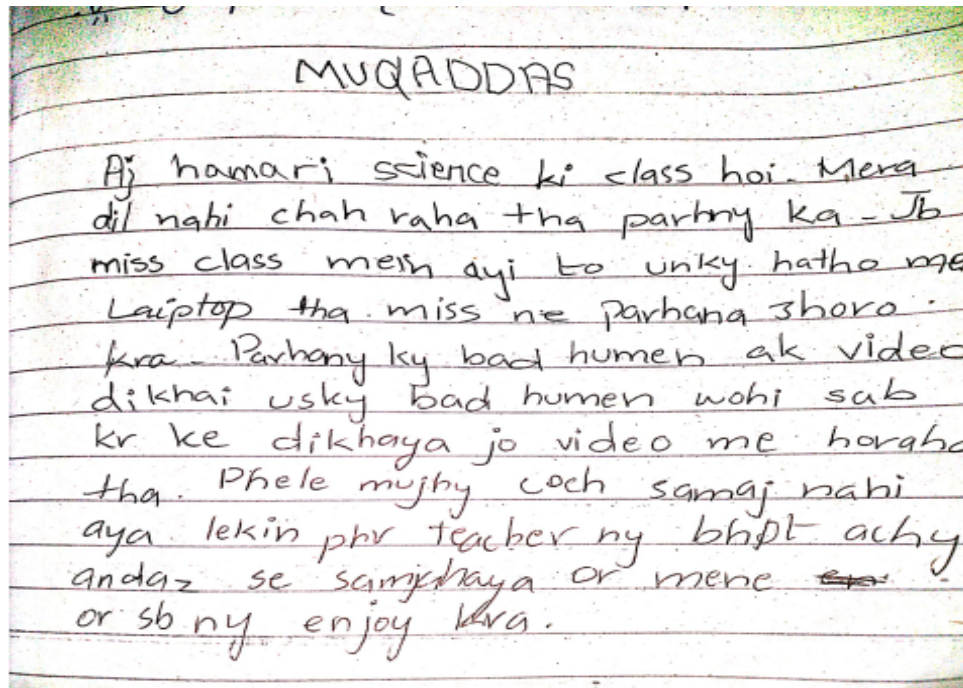
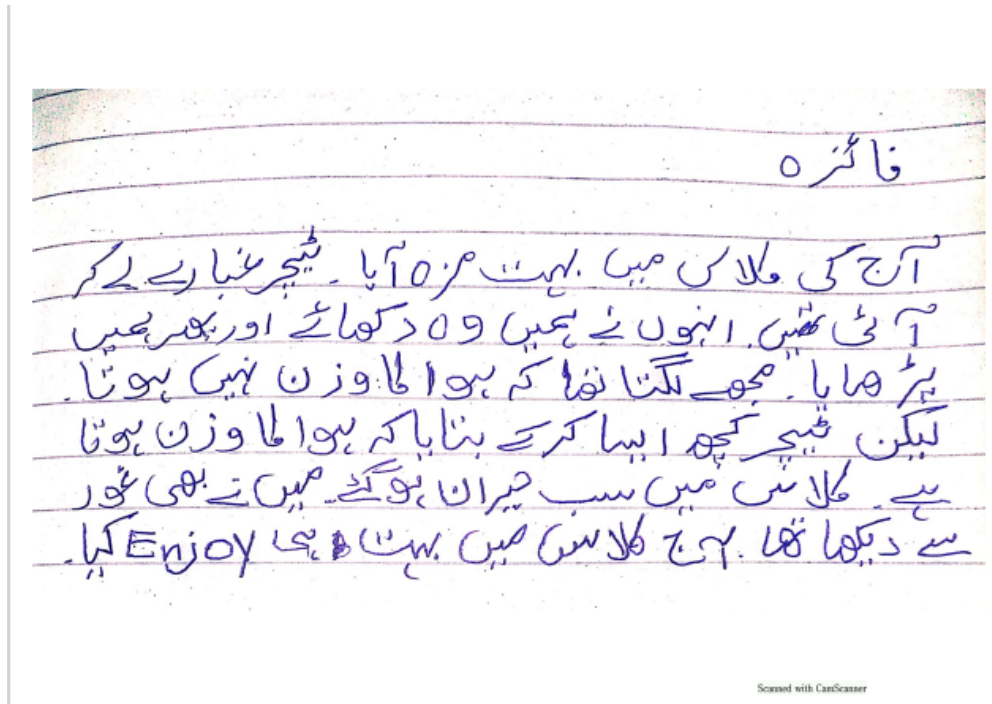
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APPENDICES

APPENDIX A

DATA COLLECTION TOOLS

A.1: Students' Reflections



A.2: Observation Checklist

Observer Information:

- Name of Observer:
- Date of Observation:
- Time of Observation:
- Classroom Grade Level:

S#	STATEMENTS	SA	A	N	D	SD
1-	Engagement level of students during the discrepant event is very high.					
2-	Questions asked by students related to the event.					
3-	There is visible signs of excitement, curiosity, or confusion among students.					
4-	Teacher introduce the discrepant event effectively.					
5-	Introduction of event generate curiosity and interest among students					
6-	Teacher employ various engagement strategies during the event (e.g., visuals, and hands-on activities)?					
7-	Percentage of students actively participating in the event.					
8-	The teacher facilitate the event very well.					
9-	Explanations are clear and understandable					
10-	The teacher encourage student inquiries and guide discussions effectively.					
11-	The event encourage students to reflect on what they've learned.					
12-	Students able to articulate their thoughts and connections made during the event.					
13-	Overall effectiveness of the teacher in implementing the discrepant event is good.					

14- Any additional observations or noteworthy incidents during the session?

A.3: Questions for Semi-Structured Interview

1. On a scale from 1 to 10, how much do you think about today's science class?
Why?
2. Have you seen in any different behavior of students that surprised you or were different from what you expected?
3. Can you recall a specific instance when I used something unexpected or surprising in a lesson?
4. Have you noticed any changes in students' interest level during the class?
What was it about those activities that captured attention?
5. What are your opinions about the science teacher who uses surprising or unexpected elements in a science lesson?
6. How do you perceive the overall engagement levels of students in science class?
7. Have you noticed any changes in student behavior or participation when incorporating discrepant events?
8. Are students able to articulate their thoughts and connections made during the event?
9. What do you think about the employment of various engagement strategies during the event (e.g., visuals, and hands-on activities)?

APPENDIX B

DETAILED LESSON PLANS

B.1: Detailed lesson plan # 1

Class: VII Date: 03rd Nov, 2023

Subject: G-Science Duration: 40 mins

Topic: Air & its Composition No. Of Students: 16

Specific Objectives: At the end of this session, students will be able to:

- 1) Define Importance of Air.
- 2) Identify composition of Air.

Teaching Resources: Chart, Pictures, Discrepant Events (Balloons, candles, Plastic Bottle, Glass of water), Activity, worksheets.

Procedure: (Initiating+ Development+ Recapitulation+ Evaluation)

Initiation:- Teacher will ask the following questions to the students which is like riddles in order to enhance curiosity in students.

- 1) What is something that you can't see but is always there? We only feel that something.
- 2) How can we breathe?
- 3) Why we need air to survive?

Here teacher will announce the topic that today we will learn about Importance of Air & its Composition.

Development:- Teacher will use Demonstration Method

Teacher will explain the importance of air by showing the related pictures & then paste them on the board.

Teacher will also give the real-world examples to the students in order to tell them the importance of air in our daily life.

Activity:- Here teacher will perform an activity

of breathing technique.

Now teacher will explain the composition of air & use blackboard for this. Making sure students learn the composition of air.

Activity :- Teacher will show some Discrepant Events to the students to make them engage in the lesson.

★ Teacher will perform 3 events.

- 1) For telling students that air occupies space.
- 2) Then for air has mass.
- 3) Then for air pressure.

During performing above mentioned events, teacher will ask some questions to students.

Recaptulation :-

Teacher will recap the important points with the help of students in class.

- 1) Importance of air
- 2) Real-world examples.
- 3) Composition of air.

Evaluation :- Teacher will give worksheets to the students.

Home-work :- Write down any three points on Imp. of air for us.

Reflection:
Supervision :-

Students are actively participating in the lesson.

Comments by Supervisor: Reflection :-

This lesson was amazing as I took all realia objects and add discrepant events as a strategy to motivate, engage students in my lesson & students enjoy all activities.

[Signature]
11/23

B.2: Detailed Lesson Plan # 2

Final Lesson
Lesson No: 30

Class: VII Date: 14-11-23
Subject: G-Science Duration: 40 min
Topic: Solutions No Of Students: 16

Specific Objectives: At the end of this session, students will be able to:
1- Define the term Solution.
2- Identify the solute & solvent present in solution.

Teaching Resources:
Colourful Pictures, Water bottle, Oil, Salt, Baking Soda, Plastic Glass, Laptop (for video), Activity-sheet, Work-sheets.

Procedure: (Initiating+ Development+ Recapitulation+ Evaluation)

Initiation:- Teacher will perform an activity in class & ask some questions to the students.
1- What is this? Water.
2- What is this? Salt
3- What happened if I mix them together? Dissolve
(Teacher will dissolve salt into water)
4- How can we specify this water? No response.
 Here teacher will announce the topic that today we will learn about Solutions.

Development:- Teacher will use Demonstration Method.
 Teacher will first explain the definition of solution to the class & write it on board.
 Then teacher will paste the chart on board and start defining some examples of solutions & components of solution.
 Then teacher will prepare some solutions by using different items like, salt, flour, soda etc to show the examples of solution & clear the

concept of solute & solvent to the students.

Activity 1:-
Teacher will distribute the activity-sheets to the students in which they perform a short activity.

Then teacher will start explaining the Water is a universal solvent by doing an experiment in class. During & after experiment teacher will continuously ask questions to the students to make sure their engagement in lesson.

Teacher will show the video on the topic water as universal solvent to the students in order to explore more information.

Activity + Recapitulation :- (Mix & Match Activity)

Teacher will give a chance to selected students to experience by their own about how water dissolves everything in itself & how solutions forms by mixing things in soil & water. & Teacher will also ask some questions to them.

Evaluation :-

Teacher will provide work-sheets to the students.

Home-Work :- Create a poster or collage showcasing different examples of solutions you use in your daily life.

Reflection:

- ① content knowledge is good
- ② Motivation is good explained very clearly.
- ③ Demonstration given very easy way.

Comments by Supervisor:

- ④ Blackboard used and explained content very clearly.
- ⑤ Model used and explained for help of content.
- ⑥ Card sheets used and explained.
- ⑦

APPENDIX C

FEEDBACK OF PRACTICUM SUPERVISOR

C.1: Practicum Supervisor's Sheet for Lesson # 1

Lesson Observation Performance for Supervisors

Class: V/ Date: 03rd Nov. 2023
 Topic: Imp. of air & its composition Subject: Science

SA	Attributes	4	3	2	1	Commentary
Lesson Planning						
1	Formulation of Objectives	✓				
2	Development of the Lesson		✓			
Personal Traits						
3	appearance	✓				
4	Voice	✓				
5	Confidence	✓				
Teaching Skills						
6	Initiation	✓				
7	Flow of Explanation	✓				
8	Content Knowledge	✓				
9	Interaction and questioning skills	✓				
10	Classroom Management	✓				
11	Use of Board	✓				
12	Use of Teaching Aids	✓				
13	Recap/Evaluation					
14	Time Management	✓				
15	Ensuring Inclusion	✓				
Total						

Any other comments or suggestions:

Signature of Observer: [Signature] 31/11/23

Meets Expectation: 4, Satisfactory 3, Needs Improvement: 2, Unsatisfactory: 1

C.2: Practicum Supervisor's Sheet for Lesson # 2

Final Lesson.

Lesson Observation Performance for Supervisors

Class: VII Date: 19-11-23
 Topic: Solutions Subject: Gr. Science.

S #	Attributes	4	3	2	1	Commentary
Lesson Planning						
1	Formulation of Objectives	✓				
2	Development of the Lesson	✓				
Personal Traits						
3	appearance	✓				
4	Voice	✓				
5	Confidence	✓				
Teaching Skills						
6	Initiation		✓			
7	Flow of Explanation	✓				
8	Content Knowledge	✓				
9	Interaction and questioning skills		✓			
10	Classroom Management		✓			
11	Use of Board	✓				
12	Use of Teaching Aids	✓				
13	Recap/Evaluation		✓			
14	Time Management		✓			
15	Ensuring Inclusion		✓			
Total						

Any other comments or suggestions:

Lesson is Successful.

Signature of Observer:
19-11-23

Meets Expectation: 4, Satisfactory 3, Needs Improvement: 2, Unsatisfactory: 1

APPENDIX D**DETAILED ACTION PLAN FOR RESEARCH****D.1: ACTION PLAN FOR CYCLE 1**

ACTIVITIES	WHEN	DATA SOURCE	DATA COLLECTION TOOL	REFLECT (What would you look for on data)
Discussion of first action plan with research supervisor	Last week of October,2023	Research supervisor	—	For planning of cycle 1
Reconnaissance: Action cycle 1 depends upon my own observation	1 st week of November,2023	My own reflections	Observation	To observed that why students of grade 6 th have lack of interest in science classroom
Action cycle 1: Different demonstration with discrepant events will be used in science classroom according to the topic	1 st week of November,2023	Students, co-operative teacher, critical friend, practicum supervisor, myself.	Reflections of students, observation sheet, and reflection of practicum supervisor	To judge the enhancement in students' engagement in science classroom

D.2: ACTION PLAN FOR CYCLE 2

ACTIVITIES	WHEN	DATA SOURCE	DATA COLLECTION TOOL	REFLECT (What would you look for on data)
Discussion of second action plan with research supervisor	2 nd week of November,2023	Research supervisor	—	For planning of cycle 2
Action cycle 2: Action cycle 2 depends upon the conclusion of action cycle	2 nd week of November,2023	Students, co-operative teacher, critical friend, practicum supervisor, & my self	Reflection of students, critical friend, co-operative teacher, practicum supervisor, semi-structured interview.	To judge the enhancement in students' engagement in science classroom
Different demonstration with discrepant events along with the technology will be used in science classroom according to the topic	2 nd week of November,2023	Students, co-operative teacher, critical friend, practicum supervisor, myself.	Reflections of students, semi-structured interview, and reflection of practicum supervisor	To judge the enhancement in students' engagement in science classroom

APPENDIX E

REFERENCE LETTER

Govt. College of Education, F.B. Area, Karachi Research Council

Reference Letter

The H.M./Principal,

Muslim Dastageer Govt. Girls Secondary School,
F.B. Area, BL-15, Karachi.

Subject: Permission to Conduct B.Ed. (Hons) Action Research Project

Dear Sir/Madam,

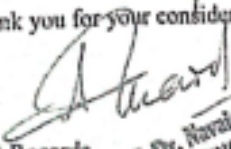
Now a days like practicum, action research is the mandatory part of B.Ed. (Hons) Curriculum.

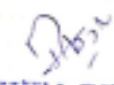
Mr./Ms. BISMA HAYAT is a bonafide student of B.Ed. (Hons) 8th semester 2023. He/She is conducting a research entitled "Using Discrepant Events in Science Classroom to improve students' engagement at Grade 6th" under the supervision of Prof. Tabassum Kausar Tabassum Kausar 11/11/23

He/She is well-versed in research ethics, and I am confident that any research conducted will be carried out with the utmost integrity and professionalism. These future teachers are enthusiastic about their students' success and constantly seek opportunities to learn and grow as an educator. Kindly allow him/her to implement innovative teaching strategies and/or to collect relevant necessary data. In this academic activity, your cooperation and facilitation will be highly appreciated.

If you have any question or further information regarding this process don't hesitate to contact our Principal or Research In-charge Dr. Sohail Ahmed (

Thank you for your consideration.

Best Regards,

Prof. Dr. Naveed Raza Siddiqui
PRINCIPAL
Govt. College of Education F.B. Area
BL-15, Karachi.


TAHIRA BEGUM
Incharge Head Mistress
Muslim Dastageer
G.G.S.S. BL-15, F.B. Area